Exact Observation Operators in Deep Unfolding for Hyperspectral and Multispectral Image Fusion

***Abstract*—Deep unfolding networks for hyperspectral and multispectral image fusion alternate data-consistency steps with learned priors. Inside these steps the observation operators are usually either learned or implemented with the zero padding of convolution layers, whereas benchmark data are simulated with a different border rule. We examine what changes when the network's operators are exactly those of the simulation, image border included. With exact operators the ground truth is a fixed point of every data-consistency step; with a zero-border or a learned operator it is not, and the step pushes it away by an amount that grows with the share of border pixels. We build PUFormer, an unfolded network that uses the exact blur–decimation and spectral operators and their exact adjoints, with a transposed-attention U-Net as prior. In an equal-budget benchmark on Chikusei and Pavia Centre, in which five published networks are retrained from their authors' code under one pipeline, PUFormer is the most accurate method on both datasets, 0.99 and 1.00 dB PSNR above the best retrained network. A module-by-module ablation on two datasets, with repeated seeds for the key rows, shows where the accuracy comes from: removing the data-consistency step costs most (0.17–0.30 dB), whereas cross-stage memory and the type of attention change PSNR by less than 0.1 dB, and a one-stage model with a third of the parameters is within 0.2 dB. At the training blur the exact operator gains only 0.08–0.15 dB over a zero border, but it makes the reconstruction agree better with its inputs and is the only operator model that can be replaced at test time: given the true blur, PUFormer loses at most 0.25 dB on either dataset for blur widths it was not trained on, whereas zero-border and learned operators lose between 0.9 and 3.4 dB, and a blur estimated from the test pair itself performs as well as the true one. Under the protocol of a recent study comparing 17 methods, PUFormer reaches 58.20 dB on Chikusei $\times$4 and 50.69 dB on Pavia Centre $\times$4. Code, training logs and all result files are public.**

***Index Terms*—Hyperspectral image fusion, multispectral image, deep unfolding, observation model, boundary conditions, super-resolution.**

I. Introduction

HYPERSPECTRAL imagers record hundreds of narrow spectral bands but, for a fixed photon budget, at a coarse spatial resolution; multispectral imagers do the opposite. Hyperspectral and multispectral image fusion recovers a high-resolution hyperspectral image (HR-HSI) from one observation of each kind [1]–[3]. Classical methods formulate fusion as a regularized inverse problem, using component substitution [4], coupled matrix factorization [5], subspace regularization [6] or fast Sylvester-equation solvers [7]. Deep networks have since become the dominant approach, from convolutional reconstruction networks [8], [9] to transformers [10]–[13] and diffusion models [14]–[17].

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Model-guided and unfolded networks [18]–[23] bring the observation model back into the network: each stage takes a step that reduces the mismatch between the current estimate and the two observations, and a learned module then refines the estimate. How faithfully the network reproduces the observation model has received less attention than the design of the learned module. In most of these networks the blur–decimation and spectral operators are learned convolution layers, or are implemented with the zero padding that convolution layers apply at the image border. The benchmark data, however, are simulated with a fixed blur and a different border rule. The network's operator is then not the one that produced the data, and even the ground truth fails the network's own consistency check. In simulated benchmarks this mismatch is avoidable, because the forward model is known exactly. A second difficulty concerns evaluation: reported numbers on the same dataset vary by tens of decibels across papers because of differences in sensor response, blur, data split, normalization and even metric definitions [3], so improvements are only meaningful within one fully specified protocol in which all methods are run.

This paper studies the operator question in a controlled way. We use an unfolded network, PUFormer, whose data-consistency steps apply exactly the simulation operators and their exact adjoints, border included, and compare it with variants that use a zero-border or a learned operator, or no data step at all, all trained with the same budget. We then compare PUFormer with published networks that we retrain ourselves under one protocol on two airborne benchmarks. Our contributions are:

1) **The role of exact operators in unfolding.** We show that with exact operators the ground truth is a fixed point of every data-consistency step, and we quantify how far a zero-border or a learned operator moves it (Proposition 1). Experimentally, the exact operator yields a small but consistent accuracy gain, a reconstruction that agrees better with its inputs, and the ability to change the blur at test time, with a known or an estimated kernel, at almost no cost.
2) **A module-by-module ablation**, with repeated seeds for the key rows, on two datasets, that identifies which parts of an unfolded network matter: the data-consistency step and the exact operator do, whereas cross-stage memory and the type of attention hardly do, and a one-stage variant with a third of the parameters loses at most 0.2 dB.
3) **An equal-budget benchmark** in which SSR-NET, PSRT, DCT, MIMFormer and the unfolded DHIF-Net are retrained from their authors' code alongside PU-

Former on Chikusei and Pavia Centre, with one data pipeline, one training budget and one metric implementation, complemented by a comparison with the 17 methods reported in [15].

4) **An analysis of the remaining errors**, showing that the Q2n shortfall on Chikusei comes from bands that the multispectral sensor does not cover and the one on Pavia Centre from dark water pixels.

Code, training logs and all result files are publicly available.

II. Related Work

A. Model-based fusion

Component substitution methods such as GSA [4] inject multispectral detail into the upsampled hyperspectral bands. Unmixing-based methods such as CNMF [5] factor both observations with shared endmembers, and subspace methods such as HySure [6] regularize the HR-HSI within a low-dimensional spectral subspace, whose dimension can be estimated with HySime [24]. FUSE [7] solves the resulting quadratic problem in closed form through a Sylvester equation. These methods use the forward model explicitly but rely on hand-crafted priors.

B. Supervised deep fusion

SSR-NET [8] and GuidedNet [9] learn a direct mapping from the two observations. Transformers model long-range spatial and spectral dependencies: PSRT [10] uses pyramid shuffle-and-reshuffle attention, MIMO-SST [11] multi-input multi-output supervision, LGCT [12] local–global collaboration, DSPNet [25] a dual spatial–spectral pyramid, SSDT [13] dilated attention, and MIMFormer [26], AMSF-Net [27], ASMNet [28] and DPFormer [29] multiscale spectral–spatial designs. Recent work also studies state-space models [30], [31], mixture-of-experts routing [32] and modality interaction [33]. General image-fusion frameworks such as U2Net [34] and OTPNet [35] have also been applied to this task.

C. Model-guided and unfolded networks

Algorithm unrolling [36]–[38] turns the iterations of an optimization method into network stages. MHF-Net [18], MoG-DCN [19] and DHIF-Net [20] unfold fusion-specific solvers, MHF-Net and DHIF-Net with operators learned as convolution layers, DUNET [21] adds transformer priors for unregistered inputs, HMPNet [22] extends unfolding to three modalities, and an information-aware transformer-based unfolding network (ITU-Net) [23], a mutual-guidance unfolding network (SMGU) [39], the perceptive spectral transformer unfolding network (PSTUN) [40], which also handles arbitrary scales, and the cross-domain-aware unfolding transformer CaFormer [41] strengthen the learned proximal steps with transformer priors. PUFormer belongs to this family, and none of its prior's components is new: passing features from one stage to the next follows memory-augmented unfolding [42], spectral-wise attention inside an unfolded network appears in DAUHST [43], and the transposed attention is that of Restormer [44]. A single unfolded network that takes the blur kernel as an input, and therefore handles kernels it was not trained on, was shown for single-image super-resolution by USRNet [45].

D. Boundary conditions

In classical deconvolution, an image border handled differently by the model and by the data causes ringing that spreads inward, and reflexive (Neumann) boundary conditions, i.e., reflect padding, are a standard remedy [46], [47]. In unfolded fusion networks the operators are either learned or implemented with the zero padding of convolution layers, while benchmark data are simulated with other border rules; to our knowledge, the effect of this mismatch has not been examined. Our contribution is on this side: an unfolded network whose operators, border included, are exactly those of the simulation, and an ablation that separates the exact operator from a zero-border and a learned operator under the same budget.

E. Generative fusion

Diffusion models have been conditioned on the two observations (DDIF [14]), constrained by the observation model during sampling (CDPS [16]), restricted to a learned subspace (ARGS-Diff [17]) or combined with differential attention in a two-stage conditioning scheme [15]. Null-space formulations [48] make the data-consistency part of the reconstruction exact. The two-stage conditional diffusion model of Chen *et al.* [15] holds the best published Chikusei $\times 4$ results in our survey and is our main reference; its comparison also includes the correlation-aware network CLSNet [49] and the dual cross transformer (DCTransformer) [50], denoted DCT in their table.

III. Observation Model and Exact Operators

Let $\mathbf{X} \in \mathbb{R}^{B\times H\times W}$ be the HR-HSI with B bands ($B = 128$ for Chikusei and 92 for Pavia Centre). The two observations are

$$\mathbf{Y}_{\mathrm{H}} = \mathcal{D}\mathbf{X}, \qquad \mathbf{Y}_{\mathrm{M}} = \mathcal{R}\mathbf{X}, \tag{1}$$

where $\mathcal{D} = \mathcal{S}\,\mathcal{K}\,\mathcal{P}$ applies, band by band, reflect padding $\mathcal{P}$ of width p, a valid correlation $\mathcal{K}$ with a normalized $(2p+1)\times(2p+1)$ Gaussian kernel, and decimation $\mathcal{S}$ that keeps every s-th pixel starting at index zero, so $\mathbf{Y}_{\mathrm{H}} \in \mathbb{R}^{B\times H/s\times W/s}$. All experiments use $s = 4$, $p = 3$ and a standard deviation of $\sigma = 2$, the 7×7 blur of [15]; we keep s and p general where the analysis does not depend on them. The spectral operator $\mathcal{R} \in \mathbb{R}^{M\times B}$ integrates the spectrum with the $M = 8$ WorldView-2 band responses [51] for Chikusei and with the $M = 4$ IKONOS band responses [52] for Pavia Centre. Because the manufacturers publish the responses only as plots together with band edges, we model each band as a flat top with logistic flanks (4 nm roll-off) that pass exactly through the published edges, at 5% response for WorldView-2, whose edges are tabulated at that level, and at 50% for the IKONOS band limits (blue 445–516, green 506–595, red 632–698 and near infrared 757–853 nm), and normalize every band to unit sum.

Exact adjoint. Every gradient step on $\|\mathcal{D}\mathbf{X}-\mathbf{Y}_\mathrm{H}\|^2$ requires $\mathcal{D}^\top = \mathcal{P}^\top\mathcal{K}^\top\mathcal{S}^\top$. $\mathcal{S}^\top$ inserts zeros and $\mathcal{K}^\top$ is a full convolution; together they are a strided transposed convolution with an output padding of $s-1$. The adjoint of reflect padding folds the padded margins back onto the pixels they were copied from. For a padded row index $j \in [0, H+2p)$ the source row is $p-j$ for $j<p$, $j-p$ in the interior and $2H-2+p-j$ in the bottom margin, hence

$$\begin{aligned}(\mathcal{P}^\top\mathbf{g})_i = \mathbf{g}_{i+p} + [1 \le i \le p]\,\mathbf{g}_{p-i} \\ + [H-1-p \le i \le H-2]\,\mathbf{g}_{2H-2+p-i},\end{aligned} \tag{2}$$

applied to rows and then to columns. With (2), $\mathcal{D}$ in the network is exactly the simulation operator, and the inner-product test $\langle\mathcal{D}\mathbf{X},\mathbf{Y}\rangle = \langle\mathbf{X},\mathcal{D}^\top\mathbf{Y}\rangle$ holds to a relative error below 10^{-13} in double precision at every scale factor. $\mathcal{R}^\top$ is the transpose of the response matrix.

Why the operator matters. Let $\tilde{\mathcal{D}}$ denote the blur–decimation operator used inside a network, and consider the data-consistency step on the LR-HSI term,

$$\mathbf{Z} = \mathbf{X}' - \alpha\,\tilde{\mathcal{D}}^\top(\tilde{\mathcal{D}}\mathbf{X}' - \mathbf{Y}_\mathrm{H}), \qquad \alpha > 0, \tag{3}$$

applied to an estimate $\mathbf{X}'$. A step of this kind should leave a correct estimate unchanged.

Proposition 1: Let $\mathbf{Y}_\mathrm{H} = \mathcal{D}\mathbf{X}$. If $\tilde{\mathcal{D}} = \mathcal{D}$, the ground truth $\mathbf{X}$ is a fixed point of (3) for every α. For any other $\tilde{\mathcal{D}}$, the step moves $\mathbf{X}$ by $\mathbf{b} = \alpha\,\tilde{\mathcal{D}}^\top(\tilde{\mathcal{D}}-\mathcal{D})\mathbf{X}$, with $\|\mathbf{b}\| \le \alpha\,\|\tilde{\mathcal{D}}\|\,\|(\tilde{\mathcal{D}}-\mathcal{D})\mathbf{X}\|$. If $\tilde{\mathcal{D}}$ differs from $\mathcal{D}$ only in its border rule (e.g., zero instead of reflect padding), $(\tilde{\mathcal{D}}-\mathcal{D})\mathbf{X}$ is nonzero only on the LR pixels whose kernel support reaches into the padding, i.e., within $\lceil p/s\rceil$ pixels of the LR border.

Proof: With $\mathbf{X}' = \mathbf{X}$ the step gives $\mathbf{Z}-\mathbf{X} = -\alpha\,\tilde{\mathcal{D}}^\top(\tilde{\mathcal{D}}\mathbf{X}-\mathcal{D}\mathbf{X}) = -\mathbf{b}$, which vanishes when $\tilde{\mathcal{D}} = \mathcal{D}$; the bound follows from the submultiplicativity of the operator norm. Both operators compute the same valid correlation of the padded image followed by the same decimation; they differ only in the padded values, which enter an output pixel only if its $(2p+1)\times(2p+1)$ support overlaps the padding, i.e., if the output pixel lies within $\lceil p/s\rceil$ pixels of the LR border. ∎

The fraction of affected LR pixels in an $h\times w$ LR image is about $2\lceil p/s\rceil(h+w)/(hw)$, so a mismatched border is a small perturbation for a 64×64 LR image but not for the 16×16 LR patches on which networks are trained. A learned operator, in contrast, differs from $\mathcal{D}$ everywhere. In both cases the learned prior has to cancel $\mathbf{b}$, a bias that depends on the image content, in addition to its actual task. The same argument explains why an exact operator can be replaced at test time: if the data were produced by a different blur $\mathcal{D}'$, using $\tilde{\mathcal{D}} = \mathcal{D}'$ restores the fixed-point property for the new data, whereas a network that has absorbed a mismatched or learned operator into its prior cannot be corrected in this way. Section VI-A measures $\mathbf{b}$ and its consequences.

IV. PUFormer

A. Unfolded iterations

PUFormer (Fig. 1) starts from the bicubic upsampling $\mathbf{X}_0$ of $\mathbf{Y}_\mathrm{H}$ and performs $K=3$ stages. Stage k first takes a gradient step on both data terms,

$$\begin{aligned}\mathbf{Z}_k = \mathbf{X}_{k-1} - s^2\alpha_k\,\mathcal{D}^\top(\mathcal{D}\mathbf{X}_{k-1} - \mathbf{Y}_\mathrm{H}) \\ - \beta_k\,\mathcal{R}^\top(\mathcal{R}\mathbf{X}_{k-1} - \mathbf{Y}_\mathrm{M}),\end{aligned} \tag{4}$$

with learnable step sizes $\alpha_k = \mathrm{softplus}(a_k)$ and $\beta_k = \mathrm{softplus}(b_k)$. The factor s^2 compensates the $1/s^2$ gain of $\mathcal{D}^\top\mathcal{D}$ on smooth content under $\times s$ decimation, so that both steps start at a comparable scale. The prior then predicts a residual,

$$\begin{aligned}(\Delta_k, \mathbf{F}_k) = \mathcal{P}_k\big([\mathbf{Z}_k, \mathbf{Y}_\mathrm{M}, s^2\mathcal{D}^\top(\mathcal{D}\mathbf{Z}_k - \mathbf{Y}_\mathrm{H})],\, \mathbf{F}_{k-1}\big), \\ \mathbf{X}_k = \mathbf{Z}_k + \Delta_k,\end{aligned} \tag{5}$$

where the third input tells the prior where the updated estimate still disagrees with $\mathbf{Y}_\mathrm{H}$, and $\mathbf{F}_{k-1}$ are the final decoder features of the previous stage (cross-stage memory, merged by a 1×1 convolution). The last convolution of every prior is initialized to zero, so training starts from the pure physics iteration. The image path, including all operator applications, stays in 32-bit floating point; only the priors run in 16-bit mixed precision [53].

Algorithm 1 PUFormer inference

Require: $\mathbf{Y}_\mathrm{H}$, $\mathbf{Y}_\mathrm{M}$, exact operators $\mathcal{D}, \mathcal{D}^\top, \mathcal{R}, \mathcal{R}^\top$
$\mathbf{X}_0 \leftarrow \mathrm{bicubic}_{\times s}(\mathbf{Y}_\mathrm{H})$, $\mathbf{F}_0 \leftarrow \varnothing$
for $k = 1, \dots, K$ **do**
  $\mathbf{Z}_k \leftarrow$ data step (4) ▷ fp32
  $\mathbf{r}_k \leftarrow s^2\,\mathcal{D}^\top(\mathcal{D}\mathbf{Z}_k - \mathbf{Y}_\mathrm{H})$
  $(\Delta_k, \mathbf{F}_k) \leftarrow \mathcal{P}_k([\mathbf{Z}_k, \mathbf{Y}_\mathrm{M}, \mathbf{r}_k], \mathbf{F}_{k-1})$ ▷ fp16
  $\mathbf{X}_k \leftarrow \mathbf{Z}_k + \Delta_k$
end for
return $\mathbf{X}_K$

B. Transposed-attention prior

Each prior is a three-level U-Net [54] with widths 48, 96 and 192, two, three and four transformer blocks on the encoder levels and bottleneck, mirrored decoder levels, and two refinement blocks. A block follows Restormer [44]: multi-Dconv-head transposed attention computes queries, keys and values with 1×1 and depth-wise 3×3 convolutions, normalizes each feature map over space, and forms a $c'\times c'$ attention matrix across channels per head (1, 2 and 4 heads on the three levels) with a learned temperature. Attention is thus computed between spectral feature maps rather than between spatial positions, its cost is linear in the number of pixels, and the receptive field of every block covers the whole image. A gated depth-wise feed-forward network with expansion 2.66 follows. The three priors hold 3.37–3.38 M parameters each; the whole network has 10.13 M parameters for Chikusei and 9.99 M for Pavia Centre, whose input and output convolutions are smaller.

C. Training objective

With ground truth $\mathbf{X}$, the loss is

$$\begin{aligned}\mathcal{L} = \|\mathbf{X}_K - \mathbf{X}\|_1 + 0.1\sum_{k<K}\|\mathbf{X}_k - \mathbf{X}\|_1 \\ + \lambda_\mathrm{SAM}\,\mathrm{SAM}(\mathbf{X}_K, \mathbf{X}) + \lambda_\mathrm{S}\big(1 - \mathrm{SSIM}(\mathbf{X}_K, \mathbf{X})\big),\end{aligned} \tag{6}$$

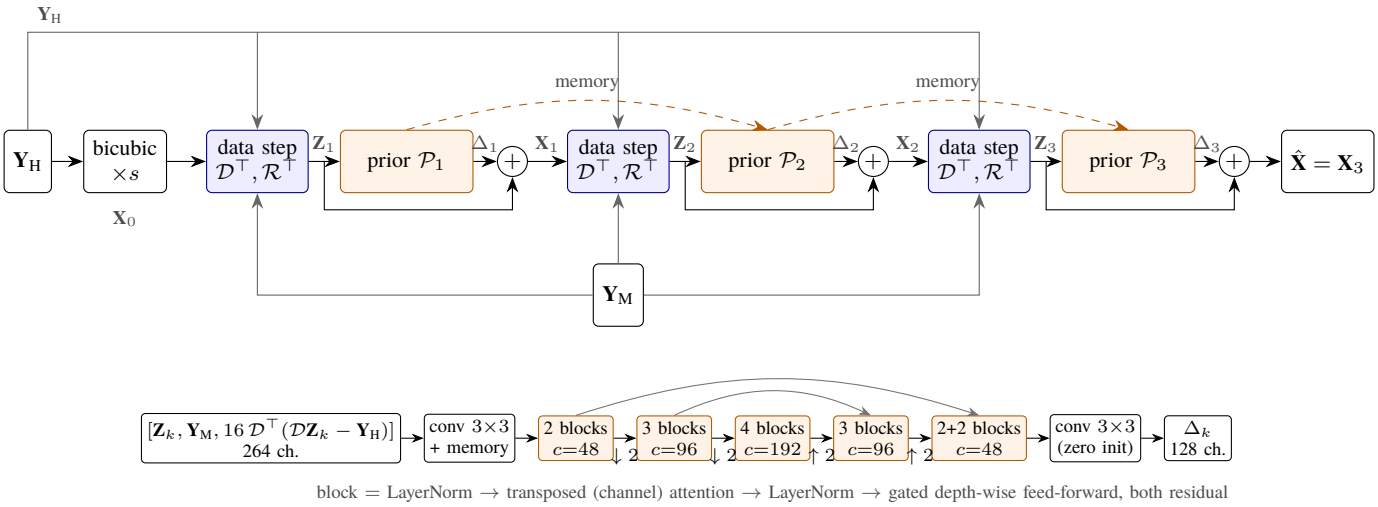


Fig. 1. PUFormer. Top: three unfolded stages, each an exact data-consistency step on both observations followed by a learned residual prior; decoder features are passed to the next stage's prior (memory). Bottom: the prior of stage k, a three-level U-Net of transposed-attention transformer blocks (channel counts for Chikusei; for Pavia Centre the input has 188 and the output 92 channels).

with mean absolute errors, deep supervision of the intermediate stages, the mean spectral angle in radians, and a band-wise Gaussian SSIM [55] whose data range is the maximum of each training patch. We use $\lambda_\mathrm{SAM} = 0.05$ and $\lambda_\mathrm{S} = 0.1$; both terms are computed in 32-bit precision. An exponential moving average of the weights (decay 0.999) [56] is used for validation and testing.

D. Self-ensemble

At test time we average the prediction for the input with the transposed prediction for the transposed input [57]. Transposition preserves the decimation phase, so both passes see observations generated by exactly the same operator. Flips and rotations do not: flipping a 256-pixel row maps sampled position $4m$ to $255 - 4m$, which has phase three, so we do not use them.

V. Experimental Setup

A. Data and protocol

Chikusei [58] is an airborne scene of 2517×2335 pixels with 128 bands from 363 to 1018 nm. The split sizes follow [15]: eight 256×256 test images, 64 validation patches and 832 training patches of 64×64. Because their locations are not specified there, we fix them as follows. We take the central 2048×2048 crop, which contains no no-data border, and divide it by its global maximum (15,133 digital numbers, DN). The test images are the eight 256×256 tiles of the top 256 rows, the validation patches tile rows 272–400, and training draws random 64×64 crops with dihedral augmentation from rows 416–2048 (800 non-overlapping patch equivalents). Sixteen-row gaps keep the three regions disjoint.

Pavia Centre [59] was recorded by the ROSIS sensor over the centre of Pavia, Italy; the distributed scene has 1096×715 pixels at 1.3 m and 102 bands between 430 and 860 nm. For this dataset, [15] uses the benchmark files released with PSRT [10]: 92 bands, a four-band multispectral image, four 256×256 test images and 142 validation and 1284 training patches of 64×64. These files are distributed through a file-sharing service that we could not access, so we rebuild the benchmark from the original scene to the same specification. We keep bands 11–102 (473–860 nm if the 102 bands are taken as evenly spaced over 430–860 nm), divide by the maximum of the retained bands (8,000 DN), apply the same blur and decimation as for Chikusei, and simulate the HR-MSI with the IKONOS responses. The test images are the four 256×256 tiles of the top-left 512×512 region. The 63 validation patches lie on a 32-pixel grid in rows 528–656, and training draws random 64×64 crops with dihedral augmentation from rows 672–1096 and from the strip to the right of the test region (rows 0–512, columns 528–715), 97 non-overlapping patch equivalents in total. Gaps of 16 pixels keep the regions disjoint. As a check on the rebuilt data, our GSA implementation reaches 34.78 dB on our test images, against 35.35 dB reported for GSA in [15].

For both datasets we follow Wald's protocol [60]: the original image serves as ground truth and both observations are simulated from it, and every image, including every training crop, is degraded on its own with (1). Table I summarizes the two protocols.

Equal-budget benchmark. Reported numbers come from different splits, normalizations and metric code (Section V-B), so for the main comparison we retrain every learned method ourselves under one pipeline: SSR-NET [8], PSRT [10], DCT [50], MIMFormer [26] and DHIF-Net [20], built from their authors' code[1]. They cover a CNN, a pyramid window transformer, the dual-cross transformer that is among the strongest entries of [15], a multiscale mixer, and a model-guided unfolding network whose blur, decimation and spec-

[1]PSRT: github.com/Deng-shangqi/PSRT (commit 9c2be90); DCT: github.com/qingma2016/DCTransformer (bd75b15); MIMFormer: github.com/meiruni/MIMFormer (0904158); DHIF-Net: github.com/TaoHuang95/DHIF-Net (6434366). Only the numbers of input and output bands are changed; DHIF-Net's code covers $\times 8$ and $\times 16$, and at $\times 4$ we use one of the two stride-4 layers of its $\times 16$ operator.

TABLE I
THE TWO $\times 4$ PROTOCOLS (DEGRADATION AND SPLIT SIZES OF [15]; IMAGE LOCATIONS OURS)

Item	Chikusei	Pavia Centre
HR-HSI	central 2048^2 crop, 128 bands	1096×715, bands 11–102 (92)
Normalization	$\div$ 15,133 DN	$\div$ 8,000 DN
LR-HSI	Gaussian 7×7, $\sigma = 2$; $\times 4$ decimation; reflect border	
HR-MSI	WorldView-2, 8 bands	IKONOS, 4 bands
Test	8 of 256^2 (rows 0–256)	4 of 256^2 (rows, columns 0–512)
Validation	64 of 64^2 (rows 272–400)	63 of 64^2 (rows 528–656)
Training	random 64^2 crops, rows 416–2048	random 64^2 crops, see text
Selection	validation PSNR only	

TABLE II
TRAINING RUNS

	Chikusei		Pavia Centre
	First run	v3 (final)	final
Initialization	random	first run	random
Border of $\mathcal{D}$	zero padding	reflect (exact)	reflect (exact)
Loss	L1 + 0.1 deep sup.	(6)	(6)
GPUs $\times$ batch	1×8	2×8	2×8
Peak learning rate	3×10^{-4}	1.5×10^{-4}	3×10^{-4}
Iterations	88,946	94,450	93,850
Training patches	0.71 M	1.51 M	1.50 M
Wall-clock time	9.5 h	10.9 h	10.2 h
Selected at	best val. (58.60 dB)	it. 44,000 (58.72 dB)	it. 43,000 (48.58 dB)

tral operators are learned convolution layers. Every method, PUFormer included, is trained for the same 4 h on one Tesla T4, with the same random patches and augmentation, batch size, EMA weights and validation-only model selection. The published networks keep their official optimizer settings (Adam, L1 loss, learning rate 10^{-4}, or 3×10^{-4} for DHIF-Net, no weight decay or gradient clipping) and are trained in 32-bit precision as released; DCT and MIMFormer use gradient checkpointing to fit into the 16 GB of a T4, which does not change what they compute. The test metrics are computed by the same code for all methods.

B. Metrics

We report the eight metrics of [15], averaged over the test images: PSNR; SSIM [55]; the spectral angle mapper (SAM, degrees, averaged over the pixels whose ground-truth and estimated spectra are both nonzero) [61]; ERGAS with ratio 4 [62]; Q2n [63] on 32×32 blocks, computed on the image in DN exactly as in the reference implementation of the pan-sharpening toolbox [64], [65]; the band-averaged correlation coefficient (CC); the spatial correlation coefficient (SCC) [66] of Sobel gradient magnitudes as in the same toolbox; and RMSE in DN. The headline PSNR is $10\log_{10}(1/\mathrm{MSE})$ over the whole normalized image, i.e., its peak is the dataset maximum. The PSNR values of [15] also use a fixed peak: for every row of their Chikusei table, $\mathrm{RMSE}\cdot 10^{\mathrm{PSNR}/20}$ is between 16.6k and 17.1k DN. Since our peak is lower (15,133 DN), our headline PSNR is conservative; with a 16.7k DN peak it would be 0.86 dB higher. For Pavia Centre the same product lies between 7.7k and 8.4k DN (median 8.07k), close to our peak of 8,000 DN, so there the PSNR values are on the same scale. The SSIM definition of [15] is not stated. We report SSIM as in the code released with PSRT [10], whose benchmark data [15] adopts (Gaussian 11×11 window, $\sigma = 1.5$, data range 1, i.e., the dataset peak), and additionally a stricter variant whose data range is each image's own maximum. We also report observation consistency, $\mathrm{PSNR}(\mathcal{D}\hat{\mathbf{X}}, \mathbf{Y}_\mathrm{H})$ and $\mathrm{PSNR}(\mathcal{R}\hat{\mathbf{X}}, \mathbf{Y}_\mathrm{M})$.

C. Implementation

PUFormer is implemented in PyTorch [67] and trained on Kaggle NVIDIA Tesla T4 GPUs (Table II). On Chikusei we train in two runs. The first run trains from scratch on one GPU with zero-padded operators and the L1 losses only. The final model (v3) fine-tunes it with the exact reflect-border operators and the full loss (6) on two GPUs with distributed data parallelism [68]. On Pavia Centre, a single run trains the model from scratch with the exact operators and the full loss on two GPUs, within one 12-hour session that also runs all tests. All runs use AdamW [69] ($\beta = (0.9, 0.99)$, weight decay 10^{-4}), a 1000-iteration linear warm-up followed by a cosine decay [70] to 10^{-3} of the peak rate that ends at the time budget, and gradient-norm clipping at 0.1. The weights with the best validation PSNR are tested once. The ablation and benchmark runs (Sections VI-A and VI-B) train from scratch with these settings, the full loss and a single GPU for 4 h each; their models are evaluated without self-ensemble.

VI. RESULTS

A. Ablation study

Table III changes one module of the full model at a time and retrains each variant from scratch with the same budget (4 h on one T4, about 40,000 iterations for the full model). On Pavia Centre the full model and the zero-border variant were trained with three seeds; their PSNR varies by only 0.02 and 0.03 dB across seeds, which sets the scale for the other differences. Chikusei repeats every variant, again with three seeds for these two rows.

Data step and operator. Removing the data-consistency step together with the residual input (row (b)) costs 0.17 dB and lowers the agreement of the reconstruction with the LR-HSI from 77.6 to 68.9 dB. Keeping the exact kernel but replacing the reflect border by zero padding (row (d)) costs 0.08 dB on average over the three seeds, and each zero-border seed is below each full-model seed. A blur kernel and spectral response learned with the network (row (e)), the operator model of DHIF-Net and MHF-Net, cost 0.06 dB. Both variants reproduce the LR-HSI about 3 dB worse than the exact model (74.5 and 73.6 against 77.6 dB). This is what Proposition 1 predicts: their data steps move the ground truth, and the prior has to compensate. On the test images the displacement $\|\mathbf{b}\|$, relative to $\|\alpha\tilde{\mathcal{D}}^\top\tilde{\mathcal{D}}\mathbf{X}\|$, is 4.66% for the zero border and 4.69% for the learned operator, and on 64×64 crops, where border pixels are a larger share, 11.11% and 11.17%. Chikusei (rows (o)–(z)) shows the same ordering with larger gaps: without the data step the PSNR is 0.30 dB lower, with a zero border 0.15 dB and with a learned operator 0.22 dB, and the agreement with the LR-HSI drops from 83.9 to 73.3, 75.9 and 77.1 dB.

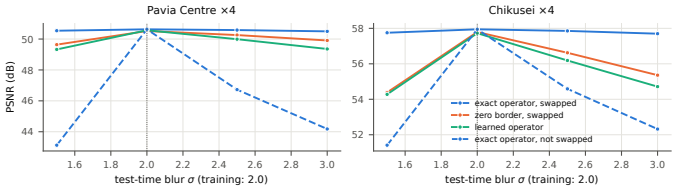


Fig. 2. Test PSNR when the blur of the test data differs from the training blur ($\sigma = 2$, dotted line), for the exact operator with and without the true blur supplied at test time, the zero-border operator with the true blur supplied, the learned operator and the model without data step.

The accuracy gap at the training blur is small; it becomes large when the blur changes (last column of Table III and Fig. 2). Given the true blur at test time, the exact model loses at most 0.13 dB for $\sigma \in \{1.5, 2.5, 3.0\}$, the zero-border model with the same swap 0.88 dB, and the learned operator, which cannot be replaced, 1.24 dB. Without the swap the exact model loses 7.50 dB: it is the most sensitive variant to a wrong operator, and the only one that fully recovers when given the right one. On Chikusei the difference is larger still: with the true blur the exact model loses at most 0.25 dB, the zero-border model 3.41 dB and the learned operator 3.45 dB. A zero border thus prevents the swap from working, even though it differs from the exact operator only at the image border, because the prior has learned to compensate a border error tied to the training blur.

Prior. Cross-stage memory (row (f)), the residual input (row (c)), the type of token mixer (rows (j) and (k)) and the number of stages (rows (g)–(i)) change the PSNR on Pavia Centre by at most 0.05 dB, comparable to the spread across seeds. A one-stage model with 3.33 M parameters is within 0.02 dB of the full model with 9.99 M; part of this parity comes from the fixed time budget, in which the smaller model completes almost three times as many iterations. The SAM and SSIM loss terms (row (l)) do not improve the PSNR, whereas deep supervision of the intermediate stages does (row (m); 0.12 dB lower without it). On Chikusei, too, removing the memory changes the PSNR by only +0.02 dB and the convolutional mixer is even slightly better (+0.08 dB), but a single stage costs 0.20 dB: on the larger, 128-band problem the extra stages are not free. We keep the full model for all other experiments, but the ablation shows that PUFormer's accuracy comes from its exact data-consistency steps rather than from its transformer prior.

B. Equal-budget benchmark

Table IV compares PUFormer with the five retrained networks; CC and SCC are listed in Table XIV. On Chikusei PUFormer reaches 57.95 dB, 0.99 dB above the best retrained network, and on Pavia Centre 50.63 dB (+1.00 dB), with 10% and 11% lower RMSE; it also has the lowest SAM and ERGAS on both datasets. MIMFormer is the strongest competitor on both. The unfolded DHIF-Net, whose operators are learned convolutions, reaches 54.39 and 47.67 dB; with 3,250 and 3,950 iterations in 4 h it is also the slowest network to train (Table XIII).

Two checks address the fairness of a fixed budget. First, the retrained baselines reach or exceed the values reported for them in [15] although our data and splits differ: on Chikusei ×4, for example, MIMFormer reaches 56.96 dB against 53.36 dB reported and PSRT 53.99 dB against 50.67 dB, so our pipeline does not weaken them. Second, the budget favors fast networks: DCT completes only 4,800 iterations in 4 h on Pavia Centre, against 37,450 for PUFormer and 97,400 for MIMFormer. Doubling the budget to 8 h for PUFormer and the strongest competitor, MIMFormer, gives **??** and **??** dB on Pavia Centre ×4, a gap of **??** dB. On Chikusei ×4, PUFormer is more accurate than the best retrained network on all test images (one-sided Wilcoxon signed-rank test, $p = 3.9e-03$).

C. Comparison with reported results on Chikusei

The comparisons in this and the next subsection place PUFormer against the 17 methods tabulated in [15]. They are less controlled than Section VI-B: the other rows are copied from [15], our test images follow its protocol but are not necessarily the same pixels, and the PUFormer models here were trained longer (Table II). Table V lists all Chikusei results of [15] together with PUFormer. The compared methods are GSA [4], CNMF [5], FUSE [7], U2Net [34], PSRT [10], DCT [50], DDIF [14], MIMFormer [26], MIMO-SST [11], OTPNet [35], SMGU [39], AMSF [27], SSDT [13], ASMNet [28], DPFormer [29], CLSNet [49] and the two-stage conditional diffusion model [15]; their values are copied from [15].

PUFormer ranks first of 18 on PSNR, SSIM (PSRT definition), SAM, ERGAS, CC, SCC and RMSE, with or without the self-ensemble. Relative to the best reported method, PSNR improves by 2.01 dB, RMSE by 26.4%, ERGAS by 12.2% and SAM by 1.1%. It ranks tenth on Q2n, which Section VI-I analyzes. Because the RMSE is expressed in sensor digital numbers, it does not depend on any choice of peak, and it gives the most direct comparison across papers. Table VI lists our own runs of reference methods under the identical pipeline and all PSNR and SSIM variants. Our GSA implementation scores 4.4 dB below the value reported in [15], so our protocol does not favor classical methods.

D. Comparison with reported results on Pavia Centre

Table VII compares PUFormer with the 17 methods of [15] on Pavia Centre. Trained from scratch in one session, PUFormer reaches 50.69 dB PSNR, 3.53 dB above the best reported value, and lowers the RMSE by 34.1% and ERGAS by 25.3%. It ranks first on six of the eight metrics (PSNR, SSIM in the PSRT definition, ERGAS, CC, SCC and RMSE), but 13th on SAM (1.883° against 1.4542°) and 16th on Q2n (0.9753 against 0.9987). With the stricter per-image data range its SSIM is 0.9970, below the reported 0.9972. The PSNR gain is larger than on Chikusei although the training region is eight times smaller (97 against 800 non-overlapping patch equivalents).

The Q2n shortfall comes from one test image, which also has the largest SAM (Table VIII, Fig. 3). On images 1, 3 and 4, Q2n is 0.9987–0.9991, at or above the best reported

TABLE III

ABLATION STUDY: EACH ROW CHANGES ONE MODULE OF THE FULL MODEL (BOLD; ✓: USED, –: REMOVED) AND IS TRAINED FROM SCRATCH WITH THE SAME 4 H BUDGET ON ONE T4 (TEST SET, SINGLE PASS). Δ: PSNR CHANGE AGAINST ROW (A) OF THE SAME DATASET. CONS.: PSNR($\mathcal{D}\hat{\mathbf{X}}, \mathbf{Y}_\mathrm{H}$)/PSNR($\mathcal{R}\hat{\mathbf{X}}, \mathbf{Y}_\mathrm{M}$) IN DB, WITH THE SIMULATION OPERATORS. SWAP: WORST TEST PSNR OVER BLUR $\sigma \in \{1.5, 2.5, 3.0\}$ WHEN THE TRUE BLUR IS GIVEN TO THE MODEL AT TEST TIME, FOR VARIANTS WITH AN EXPLICIT OPERATOR. ROWS WITH $\pm$: MEAN AND STANDARD DEVIATION OVER THREE SEEDS.

#	Data step	Res. input	Operator $\mathcal{D}, \mathcal{R}$	Memory	Mixer	K	SAM+SSIM	Deep sup.	Params (M)	PSNR	Δ	SAM	ERGAS	Cons. (dB)	Swap
Pavia Centre ×4															
(a)	✓	✓	exact	✓	MDTA	3	✓	✓	9.99	50.62±0.02	–	1.897	0.621	77.6/74.8	50.50
(b)	–	–	exact	✓	MDTA	3	✓	✓	9.99	50.46	-0.17	1.928	0.631	68.9/73.3	–
(c)	✓	–	exact	✓	MDTA	3	✓	✓	9.99	50.62	-0.01	1.900	0.621	75.7/73.8	–
(d)	✓	✓	**zero border**	✓	MDTA	3	✓	✓	9.99	50.54±0.03	-0.08	1.915	0.627	74.5/73.0	49.64
(e)	✓	✓	**learned**	✓	MDTA	3	✓	✓	9.99	50.57	-0.06	1.908	0.626	73.6/72.6	–
(f)	✓	✓	exact	–	MDTA	3	✓	✓	9.98	50.67	+0.05	1.898	0.618	77.2/74.8	–
(g)	✓	✓	exact	–	MDTA	**1**	✓	✓	3.33	50.60	-0.02	1.909	0.624	71.3/70.8	–
(h)	✓	✓	exact	✓	MDTA	**2**	✓	✓	6.66	50.64	+0.02	1.901	0.619	75.0/73.4	–
(i)	✓	✓	exact	✓	MDTA	**4**	✓	✓	13.32	50.68	+0.06	1.894	0.617	80.2/76.3	–
(j)	✓	✓	exact	✓	**conv**	3	✓	✓	9.94	50.60	-0.02	1.909	0.624	78.5/73.9	–
(k)	✓	✓	exact	✓	**window**	3	✓	✓	9.99	50.63	+0.01	1.898	0.622	78.6/75.5	–
(l)	✓	✓	exact	✓	MDTA	3	–	✓	9.99	50.66	+0.03	1.897	0.619	77.5/75.7	–
(m)	✓	✓	exact	✓	MDTA	3	✓	–	9.99	50.50	-0.12	1.900	0.629	77.7/74.6	–
Chikusei ×4															
(n)	✓	✓	exact	✓	MDTA	3	✓	✓	10.13	57.95	–	0.731	1.344	83.9/83.4	57.70
(o)	–	–	exact	✓	MDTA	3	✓	✓	10.13	57.65	-0.30	0.752	1.367	73.3/73.7	–
(p)	✓	–	exact	✓	MDTA	3	✓	✓	–	–	–	–	–	–	–
(q)	✓	✓	**zero border**	✓	MDTA	3	✓	✓	10.13	57.79	-0.15	0.738	1.357	75.9/74.4	54.38
(r)	✓	✓	**learned**	✓	MDTA	3	✓	✓	10.13	57.73	-0.22	0.739	1.362	77.1/75.3	–
(s)	✓	✓	exact	–	MDTA	3	✓	✓	10.12	57.96	+0.02	0.728	1.342	84.1/83.0	–
(t)	✓	✓	exact	–	MDTA	**1**	✓	✓	3.37	57.75	-0.20	0.743	1.359	77.4/77.1	–
(u)	✓	✓	exact	✓	MDTA	**2**	✓	✓	–	–	–	–	–	–	–
(v)	✓	✓	exact	✓	MDTA	**4**	✓	✓	–	–	–	–	–	–	–
(w)	✓	✓	exact	✓	**conv**	3	✓	✓	10.09	58.02	+0.08	0.724	1.334	85.2/85.9	–
(x)	✓	✓	exact	✓	**window**	3	✓	✓	–	–	–	–	–	–	–
(y)	✓	✓	exact	✓	MDTA	3	–	✓	–	–	–	–	–	–	–
(z)	✓	✓	exact	✓	MDTA	3	✓	–	–	–	–	–	–	–	–

TABLE IV

EQUAL-BUDGET BENCHMARK: EVERY LEARNED METHOD RETRAINED BY US UNDER THE IDENTICAL PROTOCOL AND BUDGET (4 H ON ONE T4, THE SAME DATA, PATCHES, BATCH, EMA WEIGHTS AND VALIDATION-ONLY MODEL SELECTION; SINGLE PASS). GSA AND BICUBIC INTERPOLATION NEED NO TRAINING. BEST LEARNED METHOD IN BOLD; RMSE IN DN.

	Chikusei ×4						Pavia Centre ×4					
Method	PSNR↑	SSIM↑	SAM↓	ERGAS↓	Q2n↑	RMSE↓	PSNR↑	SSIM↑	SAM↓	ERGAS↓	Q2n↑	RMSE↓
Bicubic	33.63	0.7404	4.406	9.011	0.6914	321.81	25.52	0.6282	6.554	9.852	0.6806	436.32
GSA [4]	43.20	0.9778	2.730	3.279	0.9596	107.69	34.78	0.9438	5.229	3.494	0.9276	157.93
SSR-NET [8]	–	–	–	–	–	–	–	–	–	–	–	–
PSRT [10]	53.99	0.9948	1.057	1.794	0.9887	30.57	47.87	0.9958	2.212	0.801	0.9716	32.60
DCT [50]	53.18	0.9945	1.133	2.066	0.9871	33.58	46.94	0.9953	2.442	0.888	0.9712	36.26
MIMFormer [26]	56.96	0.9964	0.801	1.488	0.9918	21.65	49.63	0.9967	1.982	0.674	0.9738	26.66
DHIF-Net [20]	54.39	0.9953	1.020	1.660	0.9900	29.37	47.67	0.9956	2.295	0.840	0.9704	33.42
PUFormer	**57.95**	**0.9969**	**0.731**	**1.344**	**0.9929**	**19.45**	**50.63**	**0.9970**	**1.897**	**0.621**	**0.9754**	**23.68**

value, but SAM is still 1.55–1.68° (mean 1.60°), above the best reported 1.454°: the water image explains the Q2n gap, but only part of the SAM gap. Image 2 contains a stretch of the river Ticino: 63% of its pixels have a positive normalized difference water index (green against near infrared), against 11, 7 and 13% in the other images, and its SAM is 2.72° and its Q2n 0.904. Water reflects little light, so its spectra are dominated by sensor noise that neither observation carries. The per-band noise that we estimate from the ground truth alone makes an angle of 3.8° with the noise-free spectrum in the darkest tenth of the pixels, against 0.23° in the brightest tenth. Over water PUFormer's SAM is 2.95°, over all other pixels 1.56°, and the darkest 10% of the pixels account for 34% of its mean SAM. The same pixels make image 2 nearly flat within many 32×32 blocks, where Q2n divides the error by a very small local variance; image 2 alone carries 97% of the Q2n deficit $\sum_i (1 - \mathrm{Q2n}_i)$ of the four images. Whether the test images of [15] contain water is not stated, so on this dataset SAM and Q2n are the least comparable of the eight metrics and RMSE the most.

As on Chikusei, our GSA scores below the value reported for it (34.78 against 35.35 dB), so the rebuilt data do not

TABLE V
CHIKUSEI ×4 COMPARISON. ALL ROWS EXCEPT PUFORMER ARE REPORTED IN [15]. BEST VALUE PER COLUMN IN BOLD. SSIM OF PUFORMER USES THE PSRT DEFINITION (DATA RANGE 1); WITH DATA RANGE EQUAL TO EACH IMAGE'S MAXIMUM IT IS 0.9969 (SINGLE PASS) AND 0.9970 (SELF-ENSEMBLE).

Method	PSNR↑	SSIM↑	SAM↓	ERGAS↓	Q2n↑	CC↑	SCC↑	RMSE (DN)↓
GSA [4]	47.5769	0.9933	1.7216	2.1667	0.9658	0.9894	0.9773	70.54
CNMF [5]	44.5291	0.9891	1.7259	2.8676	0.9662	0.9828	0.9645	101.64
FUSE [7]	41.1550	0.9787	2.5814	3.4710	0.9418	0.9736	0.9486	148.10
U2Net [34]	50.5455	0.9956	1.2389	1.8740	0.9928	0.9920	0.9798	49.97
PSRT [10]	50.6722	0.9952	1.2733	2.0107	0.9862	0.9906	0.9778	49.00
DCT [50]	52.1998	0.9977	0.9239	1.7448	0.9832	0.9935	0.9873	41.45
DDIF [14]	53.4421	0.9972	0.9387	1.8087	0.9963	0.9929	0.9840	35.58
MIMFormer [26]	53.3578	0.9973	0.9094	1.7611	0.9948	0.9931	0.9850	35.90
MIMO-SST [11]	53.6073	0.9972	0.9315	1.8119	0.9944	0.9926	0.9834	34.93
OTPNet [35]	52.8243	0.9969	1.0201	1.9332	0.9940	0.9916	0.9841	38.49
SMGU [39]	54.5244	0.9977	0.8719	1.6405	0.9952	0.9939	0.9875	31.29
AMSF [27]	49.3827	0.9944	1.4852	2.3943	0.9878	0.9879	0.9790	56.78
SSDT [13]	52.4788	0.9975	0.8768	1.7992	0.9939	0.9928	0.9847	40.00
ASMNet [28]	53.4528	0.9972	0.9472	1.9153	0.9874	0.9917	0.9848	35.89
DPFormer [29]	53.1632	0.9970	0.9483	1.8795	0.9949	0.9923	0.9824	37.18
CLSNet [49]	55.2007	0.9979	0.7836	1.6318	0.9943	0.9941	0.9869	29.06
Two-stage diffusion [15]	56.1871	0.9982	0.7203	1.5158	**0.9969**	0.9948	0.9891	25.73
PUFormer (single pass)	58.1467	**0.9989**	0.7171	1.3367	0.9930	**0.9967**	**0.9992**	19.06
PUFormer (self-ensemble)	**58.2016**	**0.9989**	**0.7126**	**1.3304**	0.9931	**0.9967**	**0.9992**	**18.94**

TABLE VI
METHODS RUN UNDER OUR PIPELINE, WITH ALL PSNR AND SSIM DEFINITIONS. P_g: PEAK = DATASET MAXIMUM (HEADLINE); P_i: PEAK = IMAGE MAXIMUM; P_b: BAND-WISE MEAN, PEAK 1. S_1: SSIM, DATA RANGE 1; S_i: DATA RANGE = IMAGE MAXIMUM.

Method	P_g	P_i	P_b	S_1	S_i	SAM	RMSE
Chikusei							
Bicubic	33.63	28.20	35.25	0.8500	0.7404	4.406	321.8
GSA [4]	43.20	37.76	44.82	0.9859	0.9778	2.730	107.7
SSR-NET [8]	49.98	44.55	53.83	–	0.9922	1.761	48.8
PUFormer, first run	57.99	52.56	65.29	0.9989	0.9969	0.728	19.3
PUFormer v3	58.15	52.71	66.26	0.9989	0.9969	0.717	19.1
PUFormer v3, ensemble	58.20	52.77	66.33	0.9989	0.9970	0.713	18.9
Pavia Centre							
Bicubic	25.52	25.11	25.87	0.6427	0.6282	6.555	436.3
GSA [4]	34.78	34.37	35.19	0.9450	0.9438	5.229	157.9
PUFormer	50.63	50.22	51.11	0.9974	0.9970	1.895	23.7
PUFormer, ensemble	50.69	50.28	51.17	0.9974	0.9970	1.883	23.5

favor our method. The reconstruction reproduces its inputs to within 79.9 dB (LR-HSI) and 77.6 dB (HR-MSI). Validation PSNR stops improving after 43,000 iterations (Fig. 4), and those weights are tested; the test PSNR, logged but never used, peaks at 50.66 dB at 25,000 iterations, against 50.63 dB for the selected weights.

E. Per-image results and visual quality

Table VIII gives the results of every test image. Chikusei images 5–8 are darker (their maximum is 0.36–0.47 of the dataset maximum), which lowers their strict SSIM but not their RMSE. Fig. 5 shows Chikusei image 7, the hardest test image, and Fig. 6 Pavia Centre image 3, the hardest there. The errors of bicubic interpolation and GSA follow every field boundary and road (mean absolute errors of 164.7 and 64.8 DN), whereas both PUFormer models keep the error small and nearly uniform over the image (11.5 and 11.0 DN), apart from a few isolated pixels.

Fig. 7 shows the error of every band. PUFormer is more accurate than GSA in all 128 bands, and the final model improves on the first run in every band, most at the red edge (700–760 nm), where the mean band RMSE falls from 7.9 to 4.2 DN (from 9.8 to 1.1 DN at 724 nm). The error is largest in the first eight bands (363–404 nm), which lie below the first WorldView-2 band and therefore receive no high-resolution detail from the multispectral image, and in the last bands near 1000 nm, where the sensor's signal-to-noise ratio drops.

F. Training stages and the exact border (Chikusei)

Table IX traces the path from the first run to the final model. Evaluating the first-run weights with the exact reflect-border operator, without retraining, drops the validation PSNR from 58.60 to 49.52 dB. The prior had learned to undo the error that the zero-padded operator introduces at the border, so with the exact operator it over-corrects there; the effect is strongest on the small 64×64 validation patches, where border pixels are a large fraction of the image. Fine-tuning removes this within 2000 iterations (58.60 dB validation) and ends at 58.72 dB. The consistency of the reconstruction with its inputs rises from 76.7 to 85.5 dB for the LR-HSI and from 75.0 to 89.0 dB for the HR-MSI. Fig. 8 shows the validation curves; validation PSNR stops improving after about 44,000 fine-tuning iterations, and the test PSNR stays between 58.13 and 58.15 dB from iteration 26,000 onward.

G. Robustness and test-time operator swap

Published methods are usually trained and tested with the same degradation. We test the final model on inputs made with a different blur or a shifted spectral response, and with Gaussian noise added to both observations (Table X, Fig. 9). Because the operators are explicit, PUFormer can also be given the true test-time operator without retraining ("swap"). With the training operator, a blur mismatch costs 3.9–7.5 dB. With

TABLE VII
PAVIA CENTRE ×4 COMPARISON. ALL ROWS EXCEPT PUFORMER ARE REPORTED IN [15] ON THE BENCHMARK DATA RELEASED WITH PSRT [10]; PUFORMER IS EVALUATED ON OUR SIMULATION OF THE SAME PROTOCOL FROM THE ORIGINAL SCENE (SECTION V). BEST VALUE PER COLUMN IN BOLD. SSIM OF PUFORMER USES THE PSRT DEFINITION (DATA RANGE 1); WITH DATA RANGE EQUAL TO EACH IMAGE'S MAXIMUM IT IS 0.9970 (SINGLE PASS) AND 0.9970 (SELF-ENSEMBLE).

Method	PSNR↑	SSIM↑	SAM↓	ERGAS↓	Q2n↑	CC↑	SCC↑	RMSE (DN)↓
GSA [4]	35.3495	0.9736	5.0812	2.7260	0.9818	0.9832	0.9768	137.78
CNMF [5]	32.1693	0.9469	3.8494	3.8446	0.9489	0.9683	0.9294	198.26
FUSE [7]	34.7418	0.9612	5.5628	3.2042	0.9103	0.9779	0.9469	146.90
U2Net [34]	44.3439	0.9963	1.6542	1.0924	0.9964	0.9973	0.9963	49.10
PSRT [10]	45.8875	0.9970	1.5230	0.9750	0.9974	0.9978	0.9966	41.85
DCT [50]	46.6745	0.9971	1.5169	0.8753	0.9973	0.9983	0.9972	37.66
DDIF [14]	46.2530	0.9970	1.5253	0.9125	0.9960	0.9982	0.9970	39.56
MIMFormer [26]	44.8497	0.9959	1.8225	1.0683	0.9957	0.9975	0.9954	46.19
MIMO-SST [11]	46.6334	0.9969	1.5587	0.8607	0.9975	0.9984	0.9972	37.68
OTPNet [35]	45.8081	0.9967	1.6573	0.9465	**0.9987**	0.9980	0.9968	41.51
SMGU [39]	47.1239	0.9972	1.4690	0.8547	0.9981	0.9984	0.9974	37.09
AMSF [27]	43.4465	0.9950	2.0533	1.2728	0.9929	0.9965	0.9940	54.25
SSDT [13]	46.3173	0.9965	1.6439	0.8918	0.9968	0.9982	0.9969	39.01
ASMNet [28]	46.2072	0.9966	1.6517	0.9019	0.9940	0.9982	0.9969	39.50
DPFormer [29]	42.4280	0.9948	1.9367	1.3986	0.9945	0.9957	0.9917	60.96
CLSNet [49]	46.5277	0.9969	1.5708	0.8412	0.9983	0.9983	0.9975	36.21
Two-stage diffusion [15]	47.1597	0.9972	**1.4542**	0.8262	0.9980	0.9985	0.9976	35.71
PUFormer (single pass)	50.6299	**0.9974**	1.8955	0.6206	0.9751	**0.9994**	**0.9995**	23.68
PUFormer (self-ensemble)	**50.6915**	**0.9974**	1.8833	**0.6170**	0.9753	**0.9994**	**0.9995**	**23.52**

TABLE VIII
PER-IMAGE RESULTS OF THE FINAL MODELS (SELF-ENSEMBLE)

Image	PSNR	S_i	S_1	SAM	ERGAS	Q2n
Chikusei, PUFormer v3						
1	59.15	0.9994	0.9994	0.644	1.550	0.9957
2	60.45	0.9985	0.9994	0.643	0.972	0.9967
3	60.97	0.9987	0.9995	0.597	1.093	0.9962
4	57.10	0.9981	0.9988	0.862	1.334	0.9917
5	57.29	0.9951	0.9986	0.726	1.304	0.9929
6	57.51	0.9957	0.9987	0.669	1.357	0.9913
7	56.24	0.9953	0.9984	0.830	1.449	0.9913
8	56.90	0.9947	0.9986	0.729	1.584	0.9889
Mean	58.20	0.9970	0.9989	0.713	1.330	0.9931
Pavia Centre, PUFormer						
1	50.48	0.9977	0.9977	1.546	0.555	0.9987
2	52.30	0.9949	0.9964	2.722	0.813	0.9044
3	49.53	0.9979	0.9979	1.677	0.557	0.9991
4	50.45	0.9975	0.9976	1.588	0.543	0.9991
Mean	50.69	0.9970	0.9974	1.883	0.617	0.9753

TABLE IX
FROM THE FIRST RUN TO THE FINAL MODEL (TEST SET UNLESS MARKED). CONS.: $\mathrm{PSNR}(\mathcal{D}\hat{\mathbf{X}}, \mathbf{Y}_\mathrm{H})$ / $\mathrm{PSNR}(\mathcal{R}\hat{\mathbf{X}}, \mathbf{Y}_\mathrm{M})$ IN DB

Model	Val.	PSNR	SAM	ERGAS	Cons.
GSA	–	43.20	2.730	3.279	45.4 / 45.0
First run (zero border)	58.60	57.99	0.728	1.356	76.7 / 75.0
+ self-ensemble	–	58.02	0.726	1.353	76.8 / 75.0
First-run weights, exact border	49.52	54.24	0.773	1.474	–
v3 (exact border, full loss)	58.72	58.15	0.717	1.337	85.5 / 89.0
+ self-ensemble	–	58.20	0.713	1.330	86.0 / 89.5

the true blur swapped in, the loss is at most 0.26 dB for σ between 1.5 and 3.0; with the zero-border operators of the first run, the same swap recovered only to 55.2–57.1 dB. A shifted spectral response is not compensated by the swap, which lowers the PSNR by a further 0.5 dB: the priors, trained with one response, rely on it beyond the data step. Noise is the main weakness of a model trained on noise-free data; PUFormer still exceeds GSA by 10.0 dB at 40 dB SNR and by 2.8 dB at 30 dB SNR. On Pavia Centre the blur behaves the same way: the training operator loses 3.3–6.8 dB and the swapped-in true operator at most 0.11 dB. Unlike on Chikusei, swapping in a shifted IKONOS response raises the PSNR slightly (by 0.2–0.3 dB), perhaps because an 8 nm shift is small relative to the 66–96 nm IKONOS bands, and the margins over GSA under noise are larger (12.8, 9.8 and 6.0 dB at 40, 35 and 30 dB SNR).

H. Operators estimated from the test pair

The swap above assumes that the test blur is known. In practice it has to be estimated. Because $\mathcal{D}$ acts on space and $\mathcal{R}$ on wavelength, they commute, and for the true blur $\mathcal{D}\mathbf{Y}_\mathrm{M} = \mathcal{R}\mathbf{Y}_\mathrm{H}$ holds exactly. We therefore estimate the blur width of each test image as $\hat{\sigma} = \arg\min_\sigma \|\mathcal{D}_\sigma \mathbf{Y}_\mathrm{M} - \mathcal{R}\mathbf{Y}_\mathrm{H}\|^2$ by a grid search with step 0.05, using only the two observations and the training spectral response, and give $\mathcal{D}_{\hat{\sigma}}$ to the network. Table XI compares this with the training operator and the true one, without and with 35 dB noise on both inputs. On Pavia Centre the estimate recovers the true width on every test image to within the grid step, also with noise (largest error 0.000), so the estimated and the true operator give the same reconstructions. In this simulated setting, where the blur is Gaussian and the spectral response known, the estimate is exact by construction; with a real sensor, a misspecified kernel shape or response would leave a residual error. With the estimate, the exact model's worst PSNR over the noise-free test blurs is 50.50 dB, against 43.13 dB with the training operator, 49.64 dB for the zero-border model with the same estimate, and 49.33 dB for the learned operator. Noise changes the picture. At 35 dB SNR all variants fall to about 44.6 dB, because none was trained with noise, and the operator models

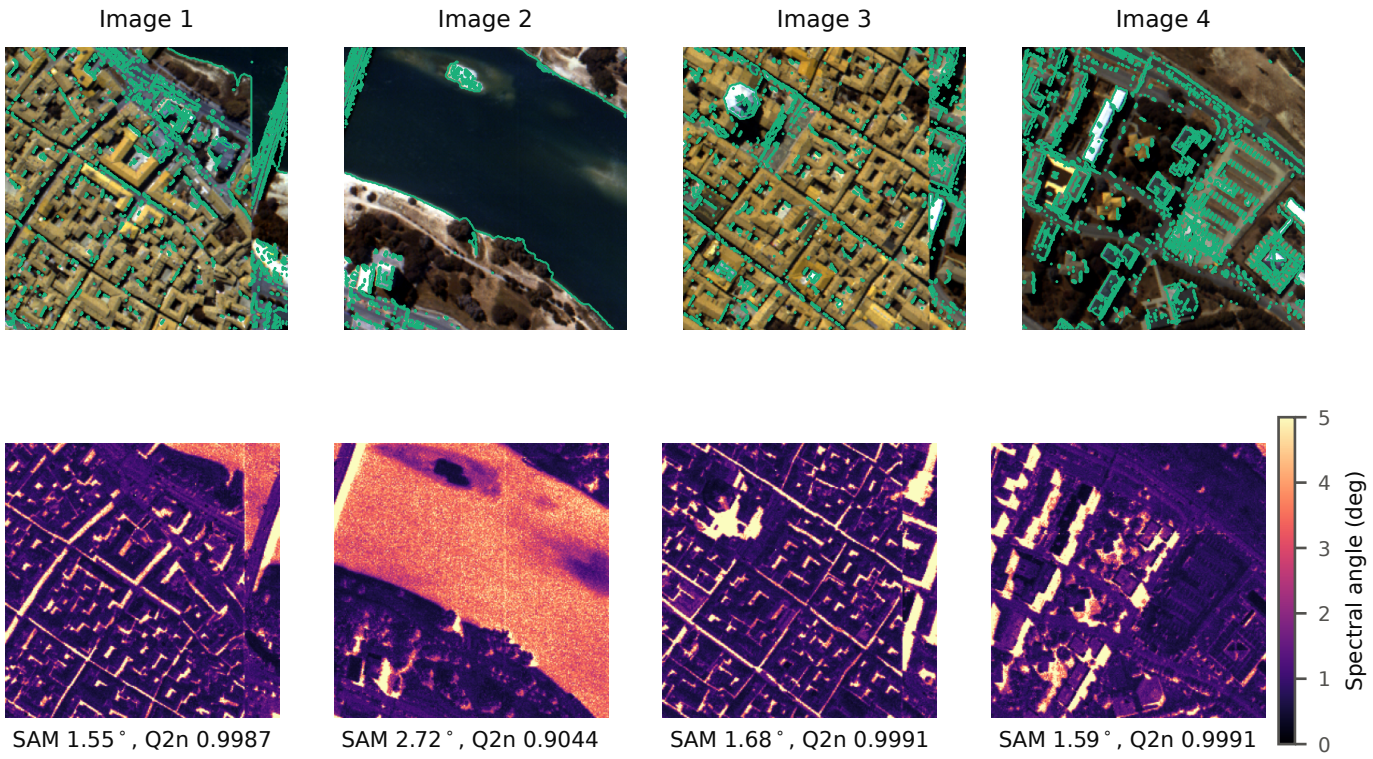


Fig. 3. The four Pavia Centre test images in false color (bands 57, 34 and 3; top), with water pixels (NDWI $>$ 0) outlined, and the spectral angle of PUFormer's reconstruction at every pixel (bottom; color scale clipped at 5°).

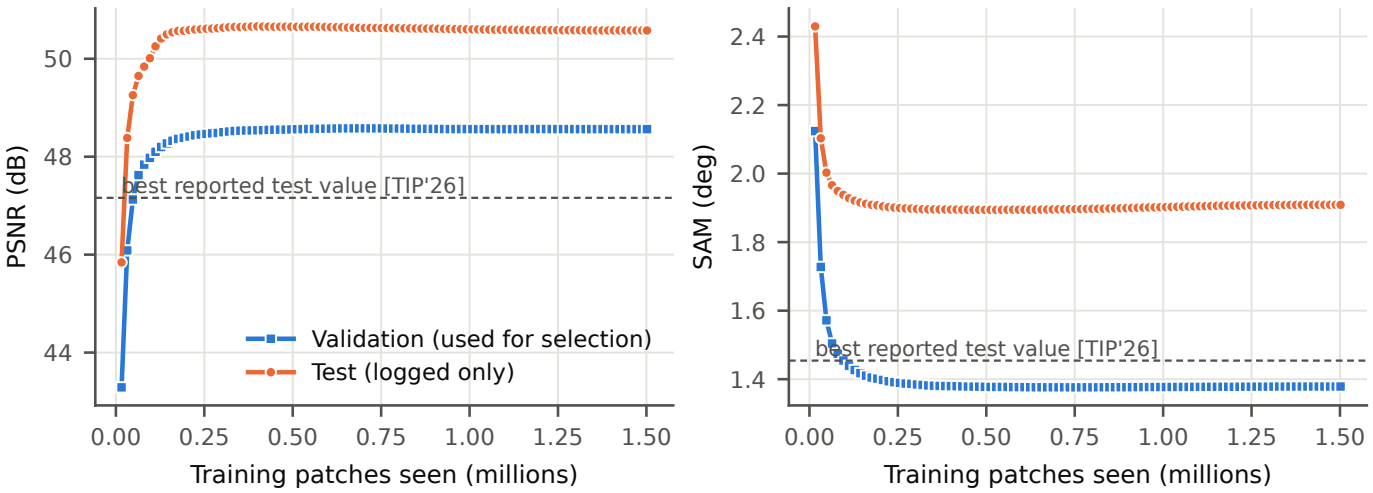


Fig. 4. Pavia Centre: PSNR and SAM during training from scratch, on the validation patches (used to select the weights) and on the test images (logged only). The dashed line is the best value reported in [15] on the PSRT benchmark data.

no longer differ: at the training blur the exact model reaches 44.56 dB, the zero-border model 44.84 dB and the learned one 44.77 dB. An exact data step enforces agreement with the observations, noise included, so its advantage requires training with noise when the inputs are noisy. The spectral response is not estimated: a swapped response did not help even when it was the true one (Table X), because the priors depend on the training response beyond the data step.

I. Where the Q2n gap on Chikusei comes from

Q2n normalizes every band of every 32×32 block by the ground truth's own mean and standard deviation in that block, so an error that is small in absolute terms counts heavily in a band that is nearly flat within a block. Table XII and Fig. 10 decompose the Q2n of the final model (0.9931). If the eight ultraviolet bands (363–404 nm) were reconstructed exactly, Q2n would rise to 0.9988; if the last 24 bands were exact as well, to 0.9998. In the worst 10% of blocks, the error energy of these eight bands equals 0.41 of the ground truth's local

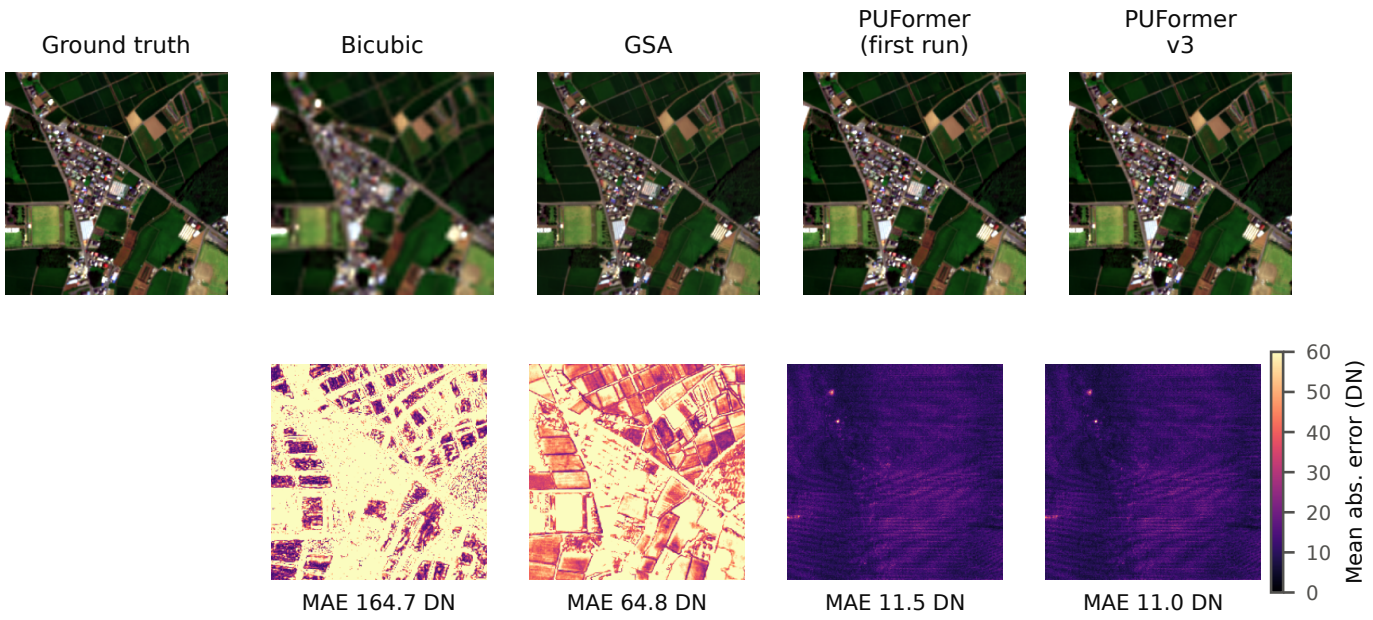


Fig. 5. Chikusei test image 7 in false color (bands 60, 40 and 20; top) and the mean absolute error over all bands in DN (bottom; color scale clipped at 60 DN).

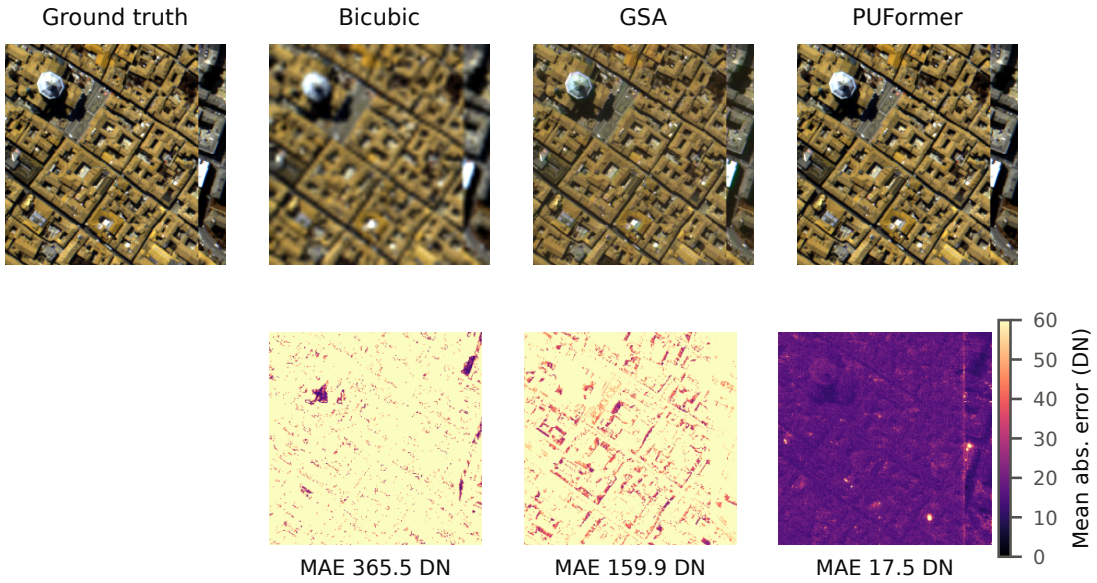


Fig. 6. Pavia Centre test image 3 in false color (bands 57, 34 and 3; top) and the mean absolute error over all bands in DN (bottom; color scale clipped at 60 DN).

variance, against at most 0.040 for any other group of eight bands. The fine detail of these bands cannot be predicted from the inputs: a least-squares injection of the multispectral detail, fitted on the test ground truth itself, reduces the band-1 error only from 61.6 to 57.5 DN, while PUFormer reaches 51.8 DN, and PUFormer is more accurate than this oracle in every band we checked. Q2n is therefore nearly insensitive to overall accuracy on these test images: blending our reconstruction toward bicubic interpolation lowers the PSNR from 58.20 to 52.15 dB but Q2n only from 0.9931 to 0.9904 (Fig. 10, right). The values reported in [15] do not follow PSNR in this way. Between 52.2 and 53.6 dB they range from 0.9832 (DCT) to 0.9963 (DDIF); of the 16 methods within the PSNR range of our curve, 11 lie above it and 5 below. U2Net, for example, is listed at 0.9928 at 50.55 dB, a value our implementation reaches only near 57.6 dB. We conclude that the Q2n values of [15] were most likely computed with a different implementation, normalization or test region, and that they are not directly comparable with ours.

J. *Complexity*

PUFormer has 10.13 M parameters and needs 18.6 GFLOPs for a 64×64 patch and 297 GFLOPs for a 256×256 image. On one Tesla T4 it reconstructs a 256×256 image with 128

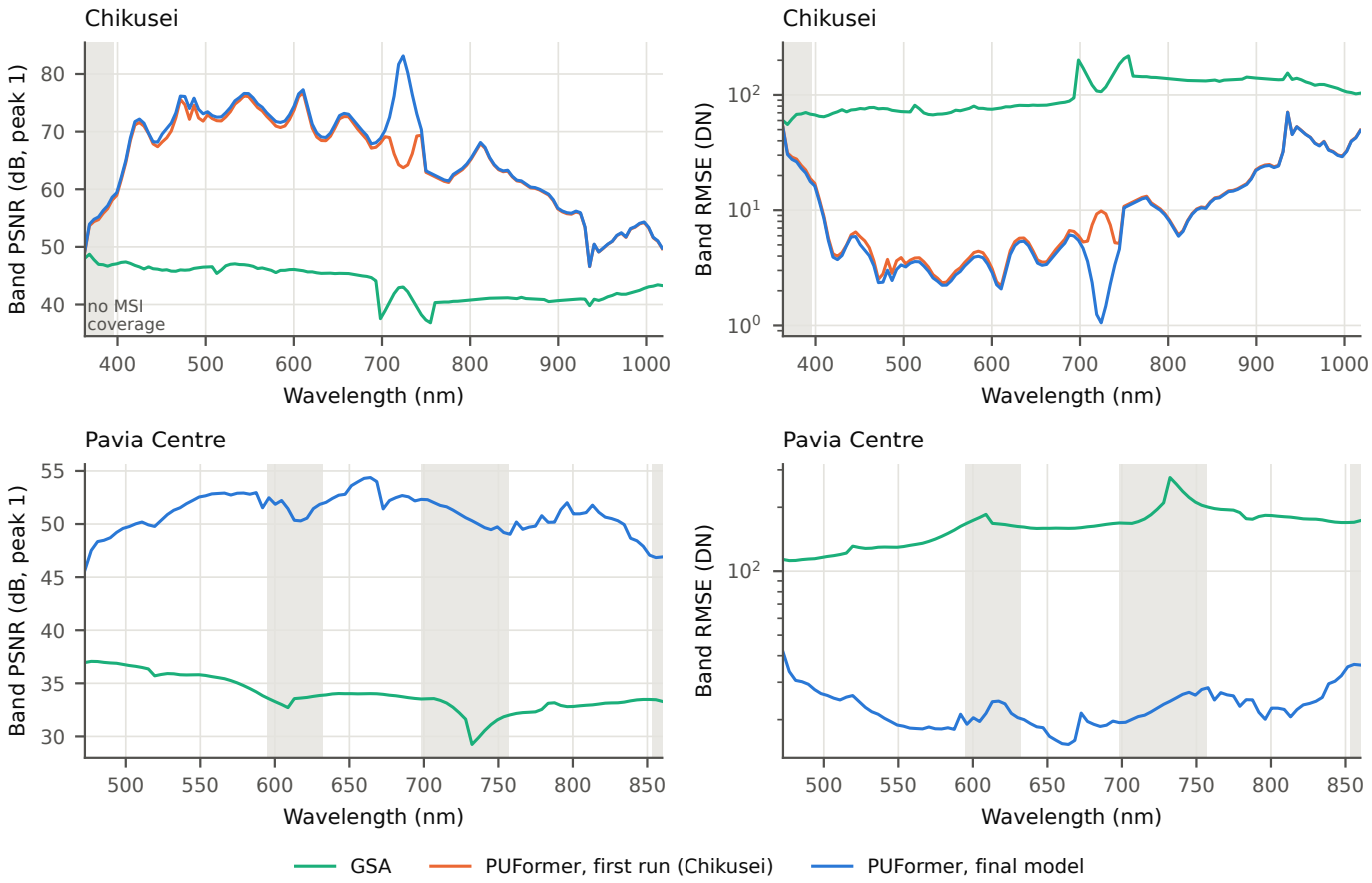


Fig. 7. Band-wise PSNR (left, peak 1) and RMSE in DN (right, log scale) over the test images of Chikusei (top) and Pavia Centre (bottom). Shaded ranges have no multispectral coverage: below the first WorldView-2 band for Chikusei, and between and beyond the IKONOS bands (50% edges) for Pavia Centre.

TABLE X
ROBUSTNESS OF THE FINAL MODELS (TEST PSNR IN DB / SAM IN DEGREES). FIXED: TRAINING OPERATOR; SWAP: TRUE OPERATOR SUPPLIED AT TEST TIME. THE BETTER PUFORMER ENTRY OF EACH CONDITION IS IN BOLD.

	Chikusei (PUFormer v3)			Pavia Centre (PUFormer)		
Condition	GSA	PUFormer fixed	PUFormer swap	GSA	PUFormer fixed	PUFormer swap
Nominal	43.20 / 2.73	58.15 / 0.72	–	34.78 / 5.23	50.63 / 1.90	–
Blur $\sigma = 1.5$	44.05 / 2.57	50.68 / 1.15	**57.97 / 0.73**	35.59 / 4.99	43.80 / 2.75	**50.56 / 1.90**
Blur $\sigma = 2.5$	42.77 / 2.82	54.26 / 0.88	**58.05 / 0.72**	34.36 / 5.36	47.30 / 2.14	**50.59 / 1.90**
Blur $\sigma = 3.0$	42.54 / 2.87	51.69 / 1.03	**57.89 / 0.74**	34.13 / 5.43	44.87 / 2.37	**50.52 / 1.91**
SRF shift -8 nm	42.86 / 2.91	**53.01 / 1.20**	52.54 / 1.31	34.58 / 5.36	49.46 / 1.99	**49.75 / 2.03**
SRF shift $+8$ nm	43.29 / 2.68	**53.03 / 1.19**	52.47 / 1.28	34.79 / 5.20	49.62 / 1.99	**49.82 / 2.02**
Noise, 40 dB SNR	43.04 / 2.85	**53.03 / 1.39**	–	34.71 / 5.37	**47.55 / 2.89**	–
Noise, 35 dB SNR	42.71 / 3.09	**49.11 / 2.20**	–	34.57 / 5.64	**44.36 / 4.19**	–
Noise, 30 dB SNR	41.84 / 3.67	**44.62 / 3.69**	–	34.16 / 6.37	**40.18 / 6.60**	–

bands in 0.20 s with mixed precision (92 bands: 0.18 s), and the self-ensemble doubles this. Because the attention is computed across channels, memory and time grow linearly with the number of pixels. Table XIII compares the parameters, operations, time and memory of all retrained networks on the same input.

VII. DISCUSSION AND LIMITATIONS

What the exact operator buys. At the blur used for training, the exact operator improves PSNR only slightly over a zero-border or a learned operator, by 0.06 to 0.22 dB. Its clearer effects are a reconstruction that reproduces its inputs more closely, and a network whose operator can be exchanged at test time, which turns a blur mismatch from a loss of several decibels into one of a few tenths of a decibel at most, with the true or an estimated blur. The ablation also shows that the transformer prior is not where the accuracy comes from: a convolutional mixer performs as well as transposed attention, cross-stage memory has no measurable effect, and a single stage costs at most 0.2 dB. We therefore see the exact operator as a property worth enforcing in any unfolded fusion network rather than as a feature of PUFormer's architecture.

Limitations. The study covers two airborne benchmarks at a scale factor of 4; other scenes, such as indoor images, and

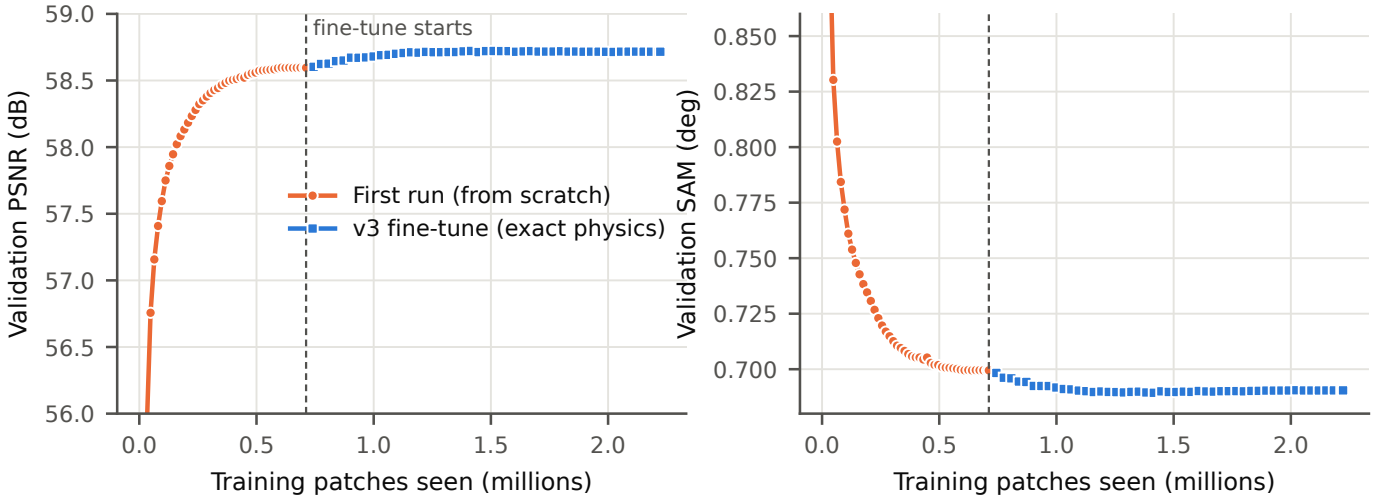


Fig. 8. Validation PSNR and SAM during training, plotted against the number of 64×64 training patches seen. The first run starts from random weights (its first two evaluations lie below the plotted range); v3 continues from its final weights with the exact border and the full loss.

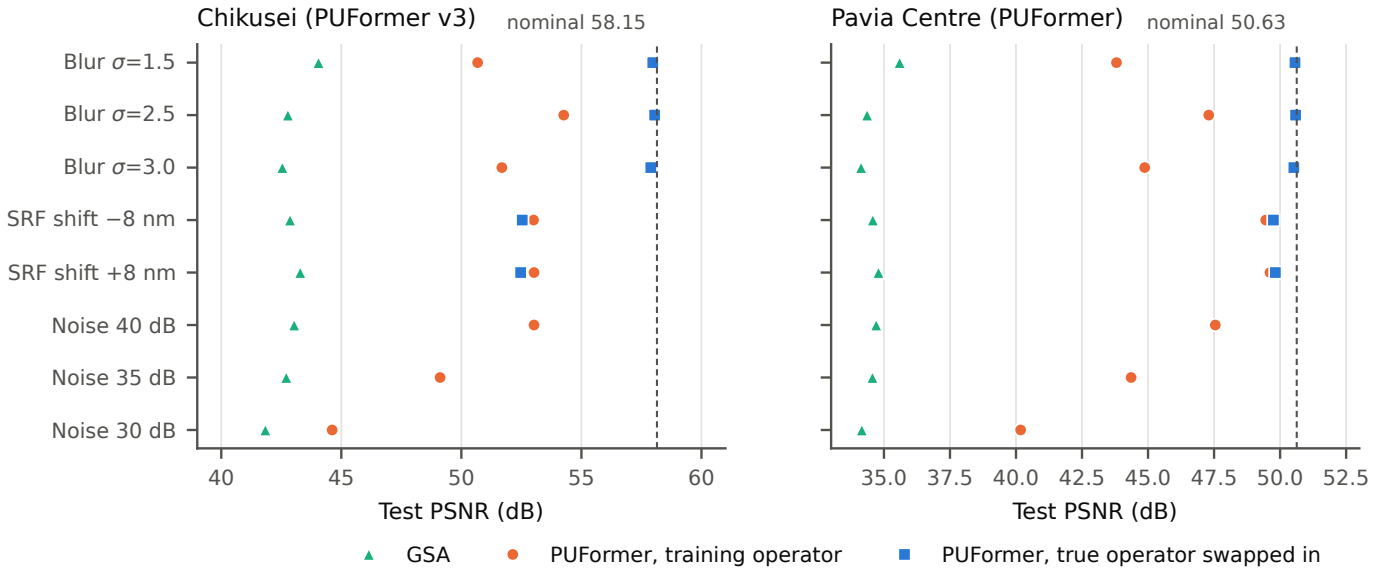


Fig. 9. Test PSNR under degradation mismatch and noise on Chikusei (left) and Pavia Centre (right). The dashed line is the nominal PUFormer result.

larger scale factors remain to be tested. All experiments use simulated data, in which the operators are known up to the parameters we vary; on real sensor pairs the blur, the spectral response and the co-registration are only approximately known. The blur estimate of Section VI-H handles the first, but a mismatched spectral response is not compensated by swapping it, and training with randomized responses would be needed. The models are trained without noise, which costs accuracy at low signal-to-noise ratios (Table X), and with noisy inputs the advantage of the exact operator disappears (Section VI-H); training with noise is needed before the operator's benefit can be expected on real data. The equal-budget benchmark gives every method the same 4 h on one GPU; this is fair in compute but favors fast networks, and although the longer-budget check and the comparison with reported values suggest that the ranking holds, longer training could narrow some gaps. The comparison with reported results in Tables V and VII rests on test images that follow the published protocol but are not necessarily the same pixels, and on Pavia Centre data rebuilt from the original scene. There RMSE in digital numbers is the most robust basis, whereas SSIM with a per-image data range, Q2n and, on Pavia Centre, SAM are below the best reported values. Finally, the test sets are small (8 and 4 images), which limits per-image significance tests.

VIII. Conclusion

We studied how exactly the observation operators inside a deep unfolding network should match the operators that generated the data. With exact operators, border included, the ground truth is a fixed point of every data-consistency step; with a zero border or a learned operator it is not. In an ablation with repeated seeds, the exact operator gives a small but consistent accuracy gain, a reconstruction that agrees better

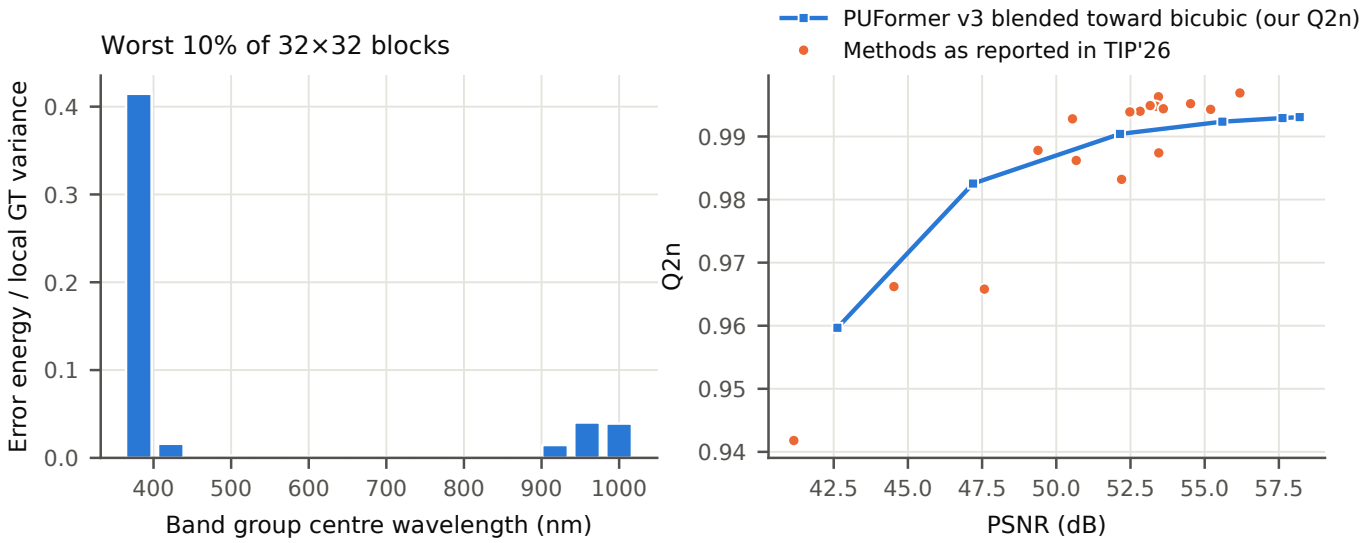


Fig. 10. Left: error energy relative to the local ground-truth variance per group of eight bands in the worst 10% of 32×32 blocks. Right: Q2n against PSNR when PUFormer's output is blended toward bicubic interpolation (our implementation), and the (PSNR, Q2n) pairs reported in [15].

TABLE XI
BLUR ESTIMATED FROM THE TEST PAIR ITSELF. $|\Delta\sigma|$: MEAN ERROR OF THE ESTIMATED BLUR WIDTH; PSNR (DB) WITH THE TRAINING OPERATOR (FIXED), THE ESTIMATED AND THE TRUE BLUR GIVEN TO THE MODEL, FOR THE EXACT OPERATOR, THE ZERO-BORDER MODEL (ESTIMATED) AND THE LEARNED OPERATOR. NOISE: 35 DB SNR ON BOTH INPUTS.

		Exact operator			Zero border	Learned
Test blur σ	$\lvert\Delta\sigma\rvert$	fixed	estimated	true	estimated	fixed
Pavia Centre ×4						
1.5	0.000	43.13	50.54	50.54	49.64	49.33
1.5 + noise	0.000	41.26	44.71	44.71	44.59	44.41
2.0	0.000	50.63	50.63	50.63	50.52	50.57
2.0 + noise	0.000	44.56	44.56	44.56	44.84	44.77
2.5	0.000	46.73	50.58	50.58	50.26	49.99
2.5 + noise	0.000	43.23	44.61	44.61	44.84	44.64
3.0	0.000	44.18	50.50	50.50	49.91	49.36
3.0 + noise	0.000	41.90	44.64	44.64	44.78	44.45
Chikusei ×4						
1.5	0.000	51.41	57.75	57.75	–	–
1.5 + noise	0.000	47.58	49.38	49.38	–	–
2.0	0.000	57.95	57.95	57.95	–	–
2.0 + noise	0.000	49.28	49.28	49.28	–	–
2.5	0.000	54.59	57.86	57.86	–	–
2.5 + noise	0.000	48.66	49.36	49.36	–	–
3.0	0.000	52.32	57.70	57.70	–	–
3.0 + noise	0.000	47.94	49.40	49.40	–	–

TABLE XII
Q2N ANALYSIS OF PUFORMER V3 (SELF-ENSEMBLE, TEST SET)

Setting	Q2n
PUFormer v3	0.9931
bands 1–8 (363–404 nm) set to ground truth	0.9988
bands 1–8 and 105–128 set to ground truth	0.9998
Deficit share, worst 10% / 50% of blocks	25% / 75%

TABLE XIII
COMPLEXITY FOR ONE 256×256 CHIKUSEI IMAGE (128 BANDS, ×4) ON ONE TESLA T4.

Method	Params (M)	GFLOPs	Time (ms)	Memory (GB)
PUFormer	10.13	296	178	0.44
SSR-NET	0.44	58	11	0.11
PSRT	0.30	43	118	0.42
DCT	8.46	2640	2103	1.10
MIMFormer	16.46	224	62	0.45
DHIF-Net	48.67	14930	2938	2.08

with its inputs, and a network whose blur can be replaced at test time with almost no loss, also when the blur is estimated from the test images. Most components of the transformer prior, in contrast, can be removed or simplified at a cost of at most 0.2 dB. In an equal-budget benchmark against five published networks retrained from their authors' code, the resulting network, PUFormer, is the most accurate on Chikusei and Pavia Centre by 0.99 and 1.00 dB PSNR, and under the protocol of [15] it reaches 58.20 dB on Chikusei ×4 and 50.69 dB on Pavia Centre ×4. Applying exact operators to real sensor pairs with estimated blur and spectral response, and extending the comparison to other scenes and scale factors, are the main directions for future work. All code, training logs and result files are publicly available.

APPENDIX A
COMPLETE BENCHMARK METRICS

Table XIV lists all eight metrics of every method in the equal-budget benchmark of Section VI-B; the per-job result files, including per-image values and training curves, are in the public repository.

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TABLE XIV
ALL METRICS OF THE EQUAL-BUDGET BENCHMARK (SINGLE PASS).

Method	PSNR	SSIM	SAM	ERGAS	CC	SCC	Q2n	RMSE
Chikusei ×4								
Bicubic	33.63	0.7404	4.406	9.011	0.8408	0.7648	0.6914	321.81
GSA	43.20	0.9778	2.730	3.279	0.9905	0.9922	0.9596	107.69
SSR-NET	–	–	–	–	–	–	–	–
PSRT	53.99	0.9948	1.057	1.794	0.9942	0.9982	0.9887	30.57
DCT	53.18	0.9945	1.133	2.066	0.9930	0.9978	0.9871	33.58
MIMFormer	56.96	0.9964	0.801	1.488	0.9960	0.9989	0.9918	21.65
DHIF-Net	54.39	0.9953	1.020	1.660	0.9951	0.9983	0.9900	29.37
PUFormer	57.95	0.9969	0.731	1.344	0.9966	0.9991	0.9929	19.45
Pavia Centre ×4								
Bicubic	25.52	0.6282	6.554	9.852	0.8174	0.7526	0.6806	436.32
GSA	34.78	0.9438	5.229	3.494	0.9958	0.9922	0.9276	157.93
SSR-NET	–	–	–	–	–	–	–	–
PSRT	47.87	0.9958	2.212	0.801	0.9990	0.9989	0.9716	32.60
DCT	46.94	0.9953	2.442	0.888	0.9988	0.9985	0.9712	36.26
MIMFormer	49.63	0.9967	1.982	0.674	0.9993	0.9993	0.9738	26.66
DHIF-Net	47.67	0.9956	2.295	0.840	0.9989	0.9988	0.9704	33.42
PUFormer	50.63	0.9970	1.897	0.621	0.9994	0.9995	0.9754	23.68

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